

Date	31 January 2025	Maximum Marks	4 Marks
Team ID	SWTID-2026-5019	Project Name	Import Data using Transform Maps (Spreadsheet)

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Architectural Specification & Stack Blueprint

ServiceNow Enterprise ETL Platform

PROJECT CONTEXT: BULK DATA MIGRATION VIA TRANSFORM MAPS

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using **Import Sets** and **Transform Maps**. It simulates a corporate scenario where bulk employee data received in Excel format is ingested into an isolated Import Set staging table (u_imp_employee_data), transformed through field mapping and coalesce logic into the live sys_user table, and audited for complete data integrity.

TECHNICAL ARCHITECTURE DESIGN GUIDELINES & DEMARCATIONS

- 1. Demarcation**

Clear boundary between Local Workstation (Excel/Browser) and ServiceNow Enterprise Cloud PaaS.
- 2. Process Logic**

Staging quarantine, Auto-Map matching, Coalesce evaluation, and onBefore transformation scripts.
- 3. Data Storage & APIs**

Temporary staging tables, target database persistence (sys_user), and REST/OOXML interfaces.

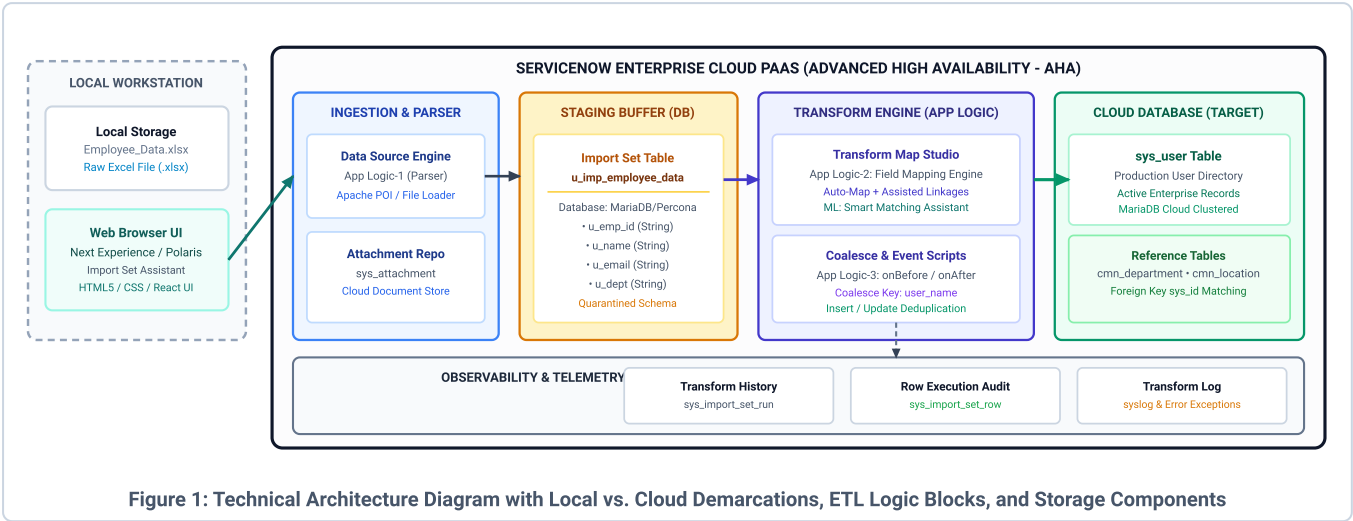


Figure 1: Technical Architecture Diagram with Local vs. Cloud Demarcations, ETL Logic Blocks, and Storage Components

Table-1: Technology Stack Components Specification

Evaluation Weight: 4 Marks

Detailed mapping of the eleven core architectural technology components governing the ServiceNow data migration pipeline:

S.No	Component	Description & Role in Solution	Technology / Specification
1.	User Interface	Administrator console for configuring Data Sources, executing mapping assist, launching transformation batches, and reviewing execution logs.	ServiceNow Next Experience (Polaris) UI, HTML5, CSS3, JavaScript, React-based Web Components
2.	Application Logic-1 (Ingestion & Parsing)	Backend process parsing spreadsheet records, reading column headers, and loading raw data into the staging table.	ServiceNow File Data Source Engine, Apache POI Spreadsheet Parser, Java Backend Worker
3.	Application Logic-2 (Transformation Engine)	Core ETL engine responsible for field-to-field matching, executing Auto Map Matching Fields, and evaluating coalesce logic.	ServiceNow Transform Engine (GlideImportSetTransformer, GlideRecord), JavaScript Runtime
4.	Application Logic-3 (Event & Validation Scripts)	Event-driven script hooks (onBefore, onAfter) sanitizing input dates, normalizing strings, and handling choice creation.	Server-side JavaScript (ES6+ / Scoped Application Runtime), ServiceNow Glide API
5.	Database (Staging Quarantine)	Temporary staging schema (u_imp_employee_data) storing unvalidated source records before production commit.	ServiceNow Import Set Table, Relational Column Store (MariaDB / Percona Clustered)
6.	Cloud Database (Target Persistence)	Enterprise system tables (sys_user) storing active employee profiles, linked with Department and Location reference tables.	ServiceNow Enterprise Cloud Database (Clustered Relational Database), InnoDB engine
7.	File Storage	Storage of source Excel spreadsheets locally on the administrator workstation and as binary attachments on the cloud instance.	Local File System (.xlsx / .csv) & ServiceNow Attachment Repository (sys_attachment)
8.	External Interface / API-1	Spreadsheet data interface reading workbook sheets, cell data types, and formatting delimiters from standard Office formats.	Microsoft Office Open XML (OOXML) Standard / MIME Type vnd.openxmlformats
9.	External Interface / API-2	Web service endpoints allowing external systems or scripts to ingest data directly into staging tables via HTTP REST.	ServiceNow Table API & Import Set REST API (JSON / HTTPS endpoints)
10.	Machine Learning / AI Model	Intelligent field matching assistant recommending source-to-target field alignments based on schema semantic similarity.	ServiceNow Predictive Intelligence & Studio Field Match Recommendation Heuristics
11.	Infrastructure (Server & Cloud Deployment)	Client-server architecture combining desktop web browsers with ServiceNow's cloud infrastructure and redundant nodes.	Local: Modern Web Browser; Cloud: ServiceNow Multi-Instance Enterprise Cloud (AHA PaaS)

S.No	Characteristics	Description & Implementation Justification	Technology / Standards Used
1.	Open-Source & Platform Frameworks	Standard open libraries and platform runtime frameworks utilized for parsing spreadsheets and executing transformation scripts.	Apache POI Engine (Excel parser), Mozilla Rhino / V8 JavaScript Engine, ServiceNow UI Framework
2.	Security Implementations	Access control enforcing Role-Based Access Control (RBAC), preventing unauthorized imports; data encrypted in transit and at rest.	TLS 1.3 Encryption in Transit, AES-256 at Rest, ServiceNow Access Control Lists (ACLs), RBAC (admin, import_admin)
3.	Scalable Architecture	Multi-instance PaaS architecture supporting batch ingestion, concurrent transform sets, and asynchronous background worker processing.	ServiceNow Concurrent Import Sets, Multi-threaded Background Schedulers, Multi-Instance Isolation
4.	Availability	Continuous operation ensured via geographically paired active-standby data centers with automatic failover and load-balanced web nodes.	Advanced High Availability (AHA) Architecture, Automated Failover Clustering, 99.99% Cloud SLA
5.	Performance	High-speed batch processing transforming 500+ records in under 60 seconds; indexed coalesce lookups and staging table cleanup routines.	Database B-Tree Indexing on Coalesce keys (user_name), Staging Table Retention Policies, Memory Caching

ARCHITECTURAL DECISION SYNTHESIS: WHY SERVICENOW NATIVE STACK?

By selecting native ServiceNow **Import Sets** and **Transform Maps** over custom scripts or third-party middleware (iPaaS), the architecture achieves an isolated staging boundary, automated coalesce deduplication, and complete audit logging natively within the platform. This eliminates external licensing costs, prevents accidental production table corruption, and provides an enterprise-grade ETL migration solution for structured spreadsheet data.

ARCHITECTURAL REFERENCES & DESIGN FRAMEWORKS

- C4 Model for Software Architecture: c4model.com
- ServiceNow System Import Sets: docs.servicenow.com
- Technical Architecture Diagrams: medium.com/the-internal-startup

- IBM Cloud Architecture Center: ibm.com/cloud/architecture
- AWS Architecture Center: aws.amazon.com/architecture
- Online Order Processing Pattern: developer.ibm.com/patterns